

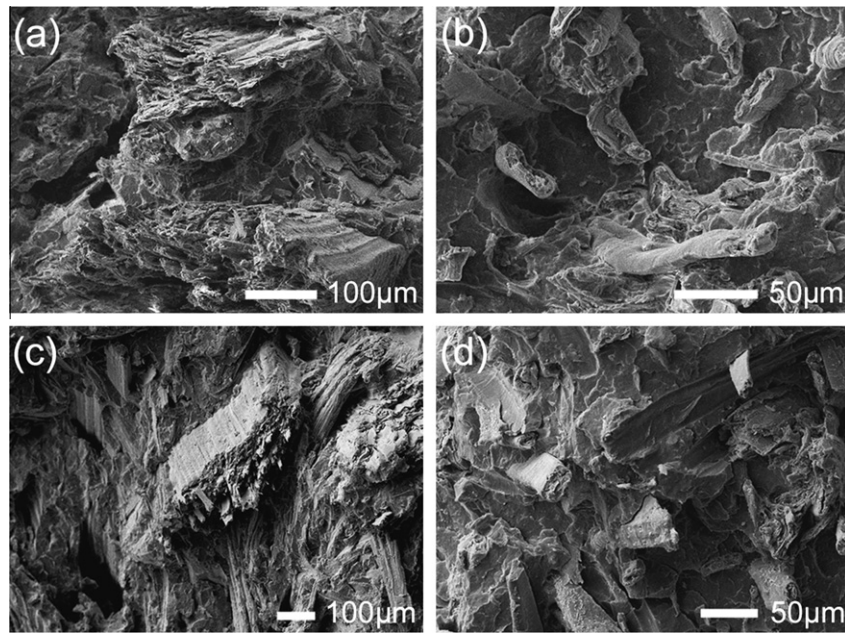


Fig. 6. Details from the fracture surfaces of the wood chip-PP composites. (a) UD-PP, (b) UU-PP, (c) PD-PP and (d) PU-PP.

Table 4

The average lengths, widths and aspect ratios of the extracted wood particles.

Material	Length, $L(l)^a$ (mm)	Width (μm)	Aspect ratio (l/d)
UD-PP	0.3	46.3	7.9
UU-PP	0.6	34.1	23.1
PD-PP	0.3	38.3	8.9
PU-PP	0.7	35.7	25.3

^a Length weighted average length.

from the length and width measurements determined using image analysis software.

As can be seen from Table 4, the wood particle samples from the composites in which dried wood chips were used (UD-PP, PD-PP) had similar aspect ratios as wood flour typically used in wood-polymer composites. The use of undried wood chips as raw material resulted in significantly greater wood particle aspect ratios after compounding. The wood chip pre-treatment slightly increased the wood particle aspect ratio, which may explain the improved mechanical properties of the composites with pre-treated wood chips. The wood particles with the highest aspect ratio were extracted from the composite made with pre-treated and undried wood chips (PU-PP), which also had the highest flexural strength and impact properties (Table 3).

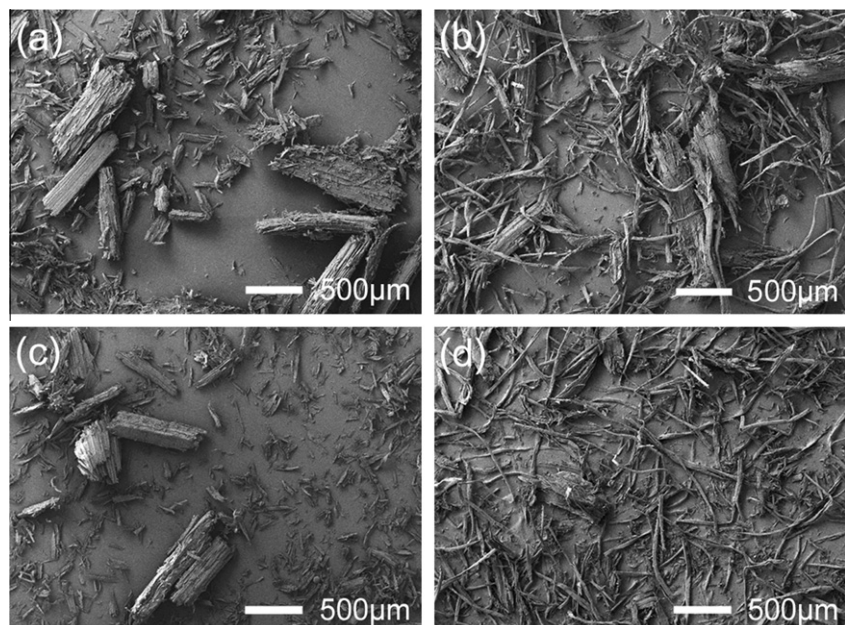


Fig. 7. Wood particles after matrix removal. (a) UD-PP, (b) UU-PP, (c) PD-PP and (d) PU-PP.

4. Conclusions

In this work the effect of chemical pre-treatment and moisture content (dried and undried wood chips) in manufacture of wood chip-PP composites was studied with an aim to create more individual wood fibres during the processing as well as improve the composites' mechanical properties. This study showed that chemical pre-treatment with sodium sulphite improved the composites' mechanical properties and that the wood particle aspect ratio was increased in the extrusion process.

The use of pre-treated wood chips enhanced the flexural properties of the wood chip-PP composites. The use of undried wood chips resulted in improved flexural strength in comparison with the composites with dried wood chips, but the flexural modulus was decreased. Study of the aspect ratios of wood particles showed that the use of pre-treated and undried wood chips resulted in the highest aspect ratio after compounding. In general, aspect ratios were considerably higher when undried wood chips were used as raw material instead of dried wood chips.

The overall results are promising. The study shows that it is possible to use wood chips as raw material for the production wood-polymer composites and that the wood chips can be defibrated during the composite manufacturing process. When a simple chemical pre-treatment is used to soften the wood a more defibrated wood structure can be achieved.

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